

MODULE 1: SENSORS AND ENERGY SYSTEM

Sensors

Sensors are basically devices which “read” a physical stimulus, and then convert that reading into an electrical signal output.

Physical Stimulus: heat, light, sound, weight, attraction

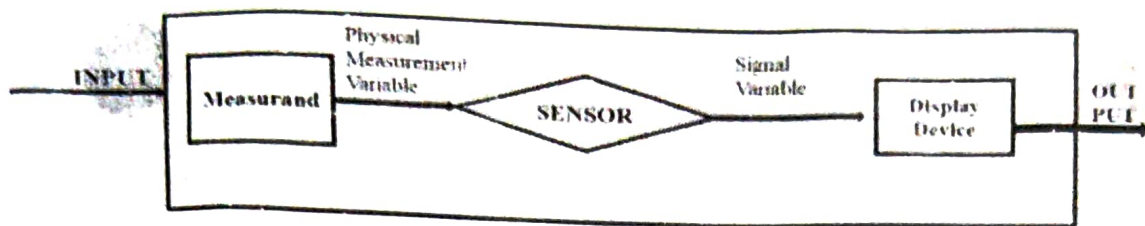


Fig: 1.1 Function of Sensor

- Sensor - observe and acquire information
- Transducer- Converts one form of energy into another form
- An actuator- Converts electrical energy into mechanical energy

Measurement process for the instrumentation model

- 1) Sensor input - the physical value or measurand (X) is observed by the sensor device
- 2) Sensor output - The sensor generates a signal variable (S) output which is normally electrical
- 3) Signal conditioning - The signal is transmitted and conditioned if needed (amplified, converted, filtered, etc.)
- 4) Display of measurement – the measurement is then displayed by the output device.

Conductometric Sensors

Conductometric sensors are two electrode devices, measures the electrical conductivity in sample solution between two electrodes.

Principle:

The basic principle of conductometric detection involves **a reaction that can change the concentration of ionic species**. This reaction leads to changes in electrical conductivity or current flow. In this method, two inert metal electrodes are used. The ions or electrons produced during an electrochemical reaction may change the conductivity or resistivity of the solution.

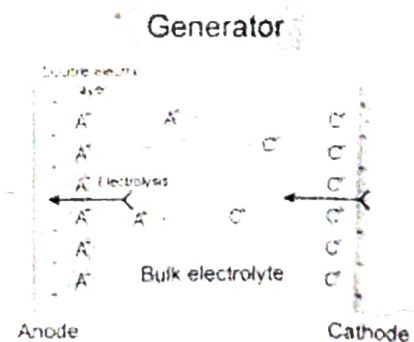
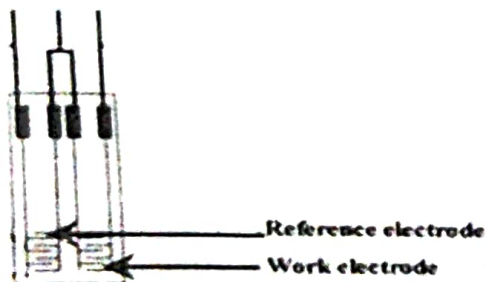


Fig. 1.5 Migration of charges in Conductometric Sensors

Working

The conductivity is result of dissociation an electrolyte, into ions. The migration of the ions is induced by an electrical field. When a potential difference is applied to the electrode, there is an electrical field within the electrolyte so the positively charged ions move towards cathode and negatively charged ions are move towards anode. (**Figure**). Thus, the current in the electrolyte is caused by the ion movement towards the electrodes where the ions are neutralized and isolated as neutral atoms (or molecules). This chemical change is recognized by working electrode and transducers converts this chemical change into electrical signal.)

Applications:

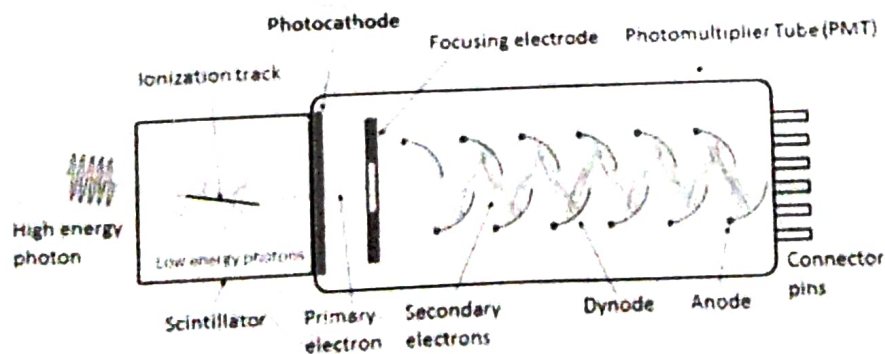
- The conductometric measuring method can be used in enzyme catalysis to determine analyte concentration and enzyme activity and selectivity.
- Screen-printed conductometric sensor with inter digital gold electrodes on glass substrate coated with molecularly imprinted polyurethane layers was fabricated to detect polycyclic aromatic hydrocarbons (PAHs) in water.
- A conductometric sensor consisting of a silicon substance with a pair of gold inter digitized and serpentine electrodes is used to determine urea.
- The conductometric biosensor based on inhibition analysis, was intended for the determination of organophosphorous pesticides.

Optical sensors

Optical sensors are electronic components designed to detect and convert incident light rays into electrical signals.

Example: Colourimetric Sensors

Principle: When a sample solution is interacted with a light of suitable wavelength, certain quantity of light is absorbed by the analyte solution and it is observed by a sensor and transducer converts intensity of absorbed light into electrical signal. The change in intensity at certain wavelength within visible (400–800nm) range can be determined using special instrumentation.



A photon (ultraviolet, visible, or near-infrared light) is converted in the photocathode into a low-energy electron, emitted into the vacuum. This electron is accelerated towards and focused onto the first dynode, releasing secondary electrons. This multiplication is repeated in subsequent dynodes, resulting in a measurable electric charge at the anode.

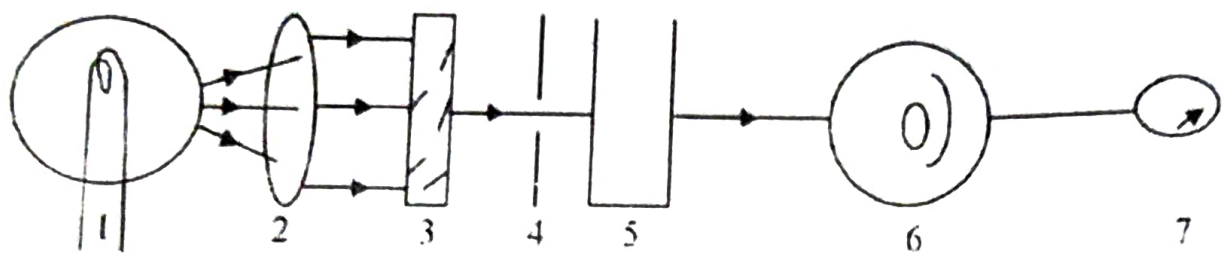


Fig. 1.6 Components of Colourimetric Sensors

A monochromatic light is made to pass through analyte solution where certain quantity of light is absorbed and it is a function of concentration of analyte. The change in the intensity of light is detected by photodetector (sensing). The light source generates an intense and stable radiation signal needed to probe an optical property of the molecular recognition element in the sensor. **The amount of absorbance is governed by Beer-lamberts law.**

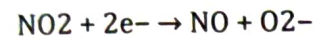
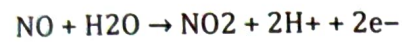
Electrochemical Gas Sensors: Principle, Construction and Applications in the Detection of NO_x and SO_x

1. Principle

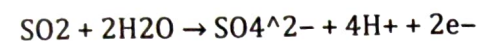
Electrochemical gas sensors operate on the principle of redox reactions occurring at an electrode surface when the target gas diffuses into the sensor. These reactions generate an electrical current proportional to the concentration of the gas

Examples of electrode reactions:

For NO_x:



For SO_x (SO₂):



2. Construction of Electrochemical Gas Sensor

Typical components:

- Working Electrode (Pt, Au, carbon)
- Counter Electrode (Pt)
- Reference Electrode (Ag/AgCl)
- Electrolyte (acidic or ionic liquid)
- Gas diffusion membrane
- Sensor housing
- Current measuring circuit

3. Working

1. Gas diffuses into the sensor.
2. Oxidation or reduction occurs at the working electrode.
3. Electrons flow through the circuit, generating measurable current.
4. Current is directly proportional to gas concentration.
4. Detection of NO_x and SO_x

Sensor for NO_x

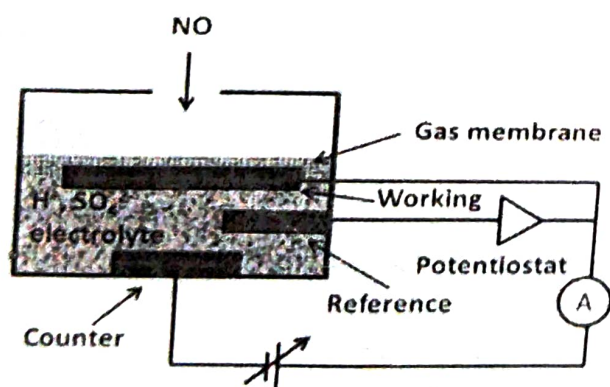


Fig: 1.14 NO_x Gas Sensor

Chemiresistive sensors based on graphene and its derivatives have been used to measure NO_x. Chemiresistive sensors measure the change in resistance upon exposure to analyte gases and can detect toxic gases at very low concentrations.

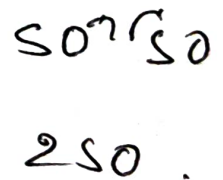
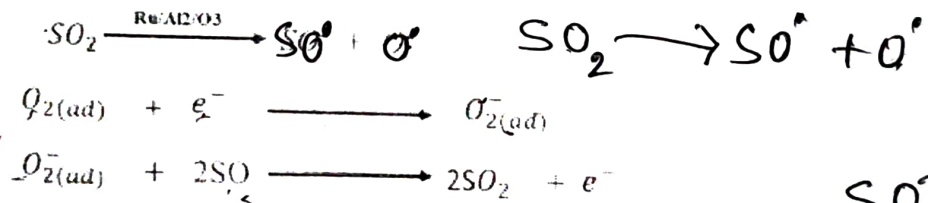
$$R_{\%} = \frac{(R_0 - R_g)}{R_0} * 100$$

Detection of SO₂: The sulfur dioxide sensor works on the electrochemical



It works based on the diffusion of SO₂ gas into the sensor. Initially SO₂ oxidizes to produce oxygen free radical and converted to molecular oxygen. Electrons are consumed from sensor for the reduction of O₂ and number of electrons decreases on the electrode surface. Resistance of the electrode is a function of Concentration of SO₂

Fig: 1.15 SO₂ Gas Sensor



$$R\% = \frac{(R_0 - R_g)}{R_0} * 100$$

R% is the reported sensor response,

R₀ is the resistance in dry clean air (background condition),

R_g is the new resistance observed under analyte gas.

Biosensor – Principle, Construction and Working for the Detection of Glucose in Biofluids

1. Principle

A biosensor is an analytical device that converts a biological response into a measurable electrical signal.

For glucose detection, enzyme-based biosensors use Glucose Oxidase (GOx).

Biochemical Reaction:

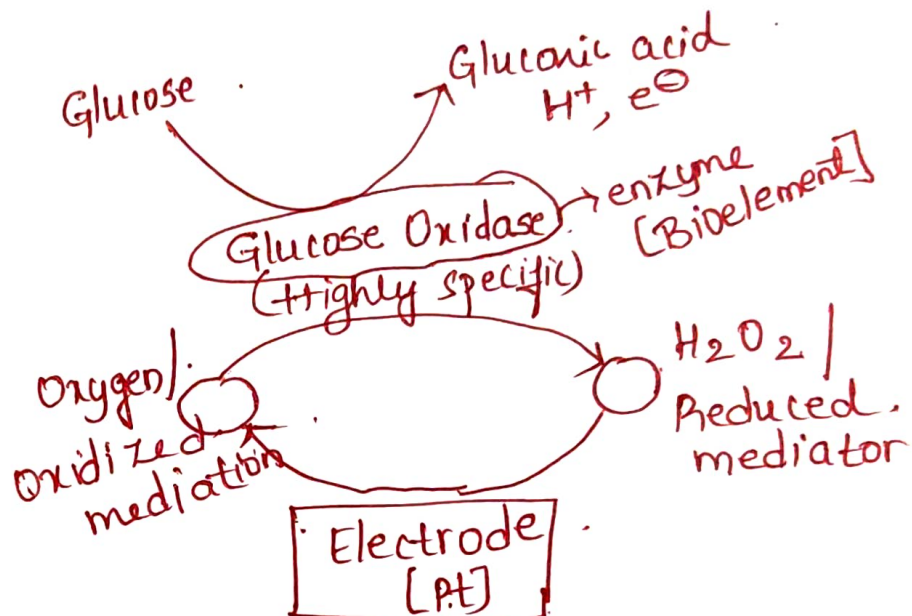


The produced hydrogen peroxide or change in oxygen level is measured electrochemically. The signal is proportional to glucose concentration.

2. Construction of a Glucose Biosensor

Components:

- Biological Recognition Element:
 - Immobilized Glucose Oxidase (GOx)
- Transducer:
 - Working electrode (Pt, Au, carbon)
 - Reference electrode (Ag/AgCl)
 - Counter electrode (Pt)
- Supporting Components:
 - Semi-permeable membrane
 - Electrolyte layer
 - Protective housing
- Signal Processing Unit:
 - Amplifier
 - ADC converter
 - Display unit



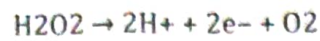
3. Working

Step 1: Sample (blood or biofluid) is introduced onto the sensor.

Step 2: Glucose diffuses to the GOx enzyme layer.

Step 3: Enzymatic oxidation produces gluconic acid and H₂O₂.

H₂O₂ oxidation:



Step 4: Generated electrons produce current measured by the transducer.

Step 5: Current is converted into glucose concentration.

Step 6: Result is displayed in mg/dL or mmol/L.

4. Applications

- Blood glucose monitoring (glucometers)
- Clinical biochemistry
- ICU continuous glucose monitoring
- Food and fermentation industry
- Glucose detection in urine, saliva, serum, interstitial fluid

5. Advantages

- High specificity
- Fast response
- Portable and easy to use
- Requires small sample volume

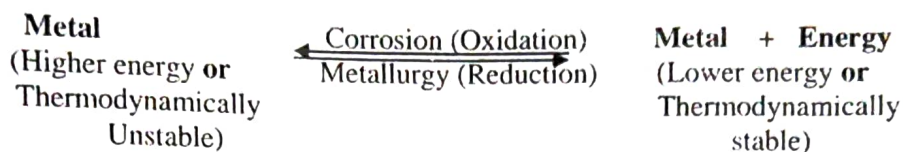
6. Limitations

- Enzyme instability
- Interference from other biomolecules
- Requires calibration

MODULE-3 CORROSION AND ELECTRODE SYSTEM

Definition of corrosion:

Corrosion is defined as "the destruction or deterioration of metals or alloys by the surrounding environment through chemical or electrochemical reactions is known as corrosion."



Examples:

- (i) **Rusting of iron:** When iron rod is exposed to the environment for a long time, a layer of reddish brown powder of Fe_2O_3 (final oxidation product) is formed on the surface of iron rod.
- (ii) **Formation of green layer on copper metal:** A green color of copper carbonate layer consisting of $[\text{CuCO}_3 + \text{Cu}(\text{OH})_2]$ is formed on the surface of copper when exposed to moist air.
- (iii) Based on the environment, corrosion is classified into (i) Dry or Chemical Corrosion (ii) Wet or Electrochemical corrosion

Difference between dry corrosion and wet corrosion:

Dry corrosion

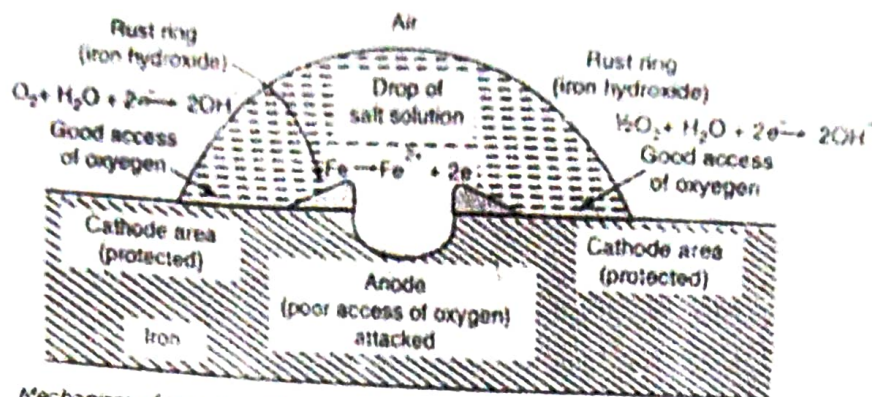
The corrosion occurs in the absence of moisture.
 It involves direct attack of chemicals on the metal surface.
 It is a slow process.
 Corrosion is uniform.
 It occurs on both heterogeneous and homogeneous surfaces.

Wet corrosion

The corrosion occurs in the presence of moisture or conducting medium.
 It involves formation of galvanic cells.
 It is a fast process.
 Corrosion is not uniform. It depends on the size of anodic part of metal.
 It occurs only on heterogeneous metal surfaces.

Electrochemical theory of corrosion:

When a metal such as iron is exposed to atmospheric air for long time, the following electrochemical reactions occur gradually on the surface of iron metal.

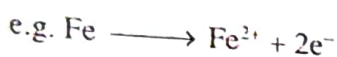


Mechanism of wet corrosion by oxygen adsorption (rusting of iron)

Fig: 3.1 Rusting of Iron mechanism

Reaction at anodic region

As mentioned above, oxidation of metal takes place at anode.



Reactions at cathodic region

At cathode, the reaction is either

(a) Liberation of hydrogen

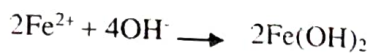
or

(b) Absorption of oxygen.

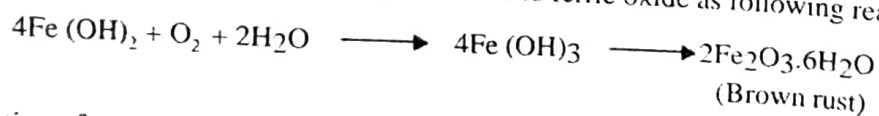
Liberation of hydrogen (in the absence of oxygen)	Absorption of oxygen (in the presence of oxygen)
In acidic medium the reaction is $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\uparrow$ In neutral or alkaline medium, the reaction is $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^- + \text{H}_2\uparrow$	In acidic medium, the reaction is $4\text{H}^+ + \text{O}_2 + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$ In neutral or alkaline medium, the reaction is $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$

Overall reaction

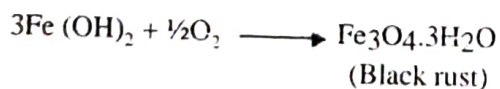
The metal ions (Fe^{2+}) liberated at anode and some anions (OH^-) formed at cathode diffuse towards each other through the conducting medium and form a corrosion product somewhere between the anode and cathode as



In an oxidizing environment, the insoluble $\text{Fe}(\text{OH})_2$ oxidized to ferric oxide as following reaction.



If the concentration of oxygen is limited then $\text{Fe}(\text{OH})_2$ is converted into magnetic oxide of Fe and is known as black rust.



Factors affecting the rate of corrosion:

The rate of corrosion is mainly dependent upon environmental conditions. The following are the factors which decide the rate of corrosion.

i). Primary factors (Related to nature metal)

a. Ratio of anodic to cathodic areas: The rate of the corrosion is mainly depends on relative sizes of anodic and cathodic areas. If the metal has smaller the anodic area and larger the cathodic area results, intense and faster corrosion takes place at the anodic area. This process makes the anodic reaction to takes place at maximum rate thus increasing the corrosion rate.



Fig: 3.2 Ratio of anodic to cathodic areas on the rate of corrosion

Example: coating of tin (Sn) on iron (Fe) surface: in this case if coating is not uniform and some areas are not covered, a small anode and a large cathode areas are formed. An intense corrosion occurs at the anodic area.

On the other hand, coating of Zn on Fe sheet gives an anodic coating to iron. Even if zinc coating peels off at some areas the rate of corrosion of iron is low. This is because of the formation of large anodic and small cathodic areas (Fig.3.2).

b. Nature of corrosion product

Usually product of corrosion is metal oxide. The metal oxide covers the metal and form a thin metallic layer. This metallic layer determines the speed of corrosion. If the corrosion product is highly ionic, porous, conducting, non-stoichiometric & unstable then, the rate of corrosion increase. On the other hand, if corrosion product is less ionic, nonporous, non-conducting, stoichiometric and stable then the corrosion product deposited on the metal surface prevents further corrosion.

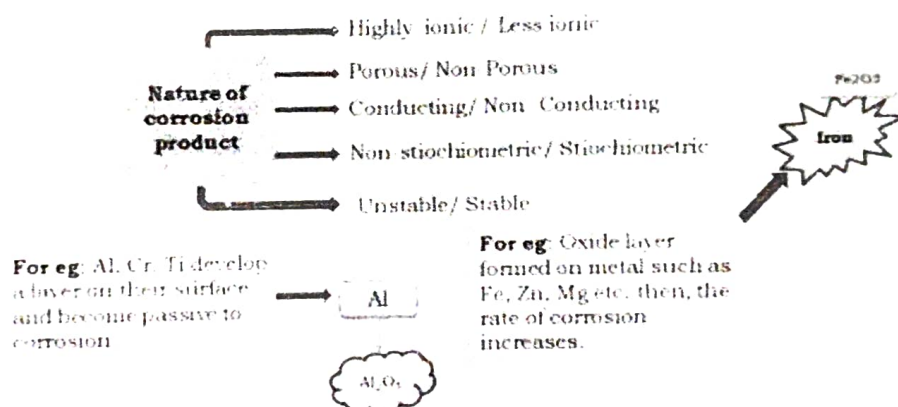


Fig.3.3 Nature of corrosion product and rate of corrosion

ii). Secondary factors (Related to nature of the medium)

a.pH of the medium:

In general, the rate of corrosion increases with decrease in pH. In case of metal such as iron, at very high pH (pH above 10) protective layer of iron oxide is formed which prevents further corrosion. Between pH 3 and 10, in the presence of oxygen is essential for corrosion of iron. But, if the pH of the medium is 3 or lower than 3 severe corrosion occurs even in the absence of air, due to the continuous evolution of H_2 gas at cathode.

b. Temperature

The rate of corrosion is directly proportional to temperature i.e., rise in temperature, increases the rate of corrosion. This is because the rate of diffusion of ions increases with rise in temperature. In some cases the rise in temperature decreases the passivity leading to an increase in the corrosion rate. Example: Steel and other metals corrode at faster rates at higher temperatures than at lower temperatures.

a. Conductance of the medium:

The corrosion rate increases in the presence of conducting species in the corroding environment. The speed of corrosion in dry medium is less than that in a wet medium. Because wet medium contains more conducting species compared to wet medium.

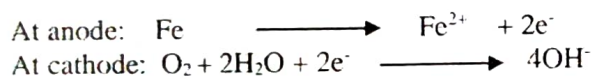
Types of corrosion:

1. Differential metal corrosion: Differential metal corrosion arises when two dissimilar metals such as iron and copper are in contact with each other in a corrosive medium. The potential difference between two metals is the cause or driving force for corrosion. The metal with lower electrode potential or more active metal (top in electrochemical series) acts as anode and undergoes corrosion quickly. Other metal (cathode) is unaffected from the corrosion. Example: In Fe-Cu galvanic cell, iron ($E^\circ = -0.44V$) with lower reduction potential than copper ($E^\circ = +0.34V$), hence zinc acts as anode and the electrons flow from anodic metal to cathodic (Cu) Fig.2.3.



Fig: 3.4 Galvanic Corrosion

In corrosive medium,



Factors affecting rate of galvanic corrosion:

- Higher the potential difference between the anodic and cathodic metals, higher is the rate of corrosion.
- Smaller anodic area and larger cathodic area increases the rate of galvanic corrosion.

This type of corrosion can be observed in the following examples:

- Steel screw's in a brass marine hardware.
- Steel pipe connected to copper plumbing.
- Zinc coated mild steel structures.

2. Differential aeration corrosion

This type of corrosion occurs when two different parts of the same metal are exposed to different oxygen concentrations.

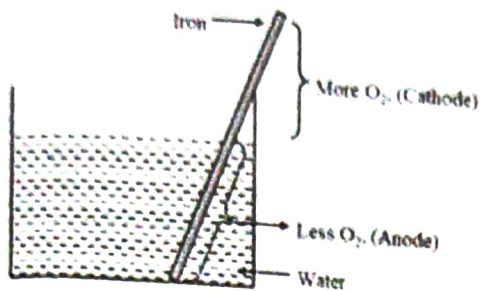


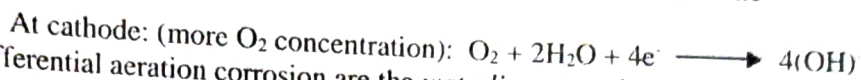
Fig: 3.5 Differential aeration corrosion

(e.g. An iron rod partially dipped in water.) The part of the metal which is exposed to less oxygen concentration acts as anode as shown in Fig:3.5. The part which is exposed to more oxygen concentration acts as cathode. Thus, an oxygen concentration cell is formed.

Metal/Less oxygen/more oxygen/Metal

The anodic region undergoes corrosion and the cathodic region is unaffected.

The reactions may be represented as follows:



Examples of differential aeration corrosion are the *waterline corrosion* and *pitting corrosion*.

2. a. **Water line corrosion:** This is an example of differential aeration corrosion.

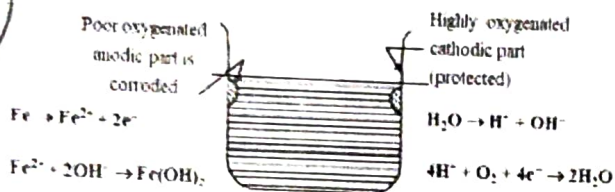
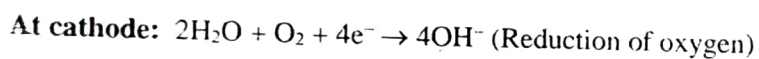
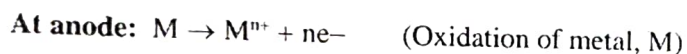


Fig: 3. 6 Waterline Corrosion

When a steel tank is partially filled with water for a long time, the inner portion of the tank below the water line is exposed only to DO , whereas, the portion above the water line is exposed to more oxygen. Thus the portion below the water line acts as anode and undergoes corrosion. The upper portion acts as cathode and is unaffected. A distinct brown line is formed just below the water line due to the deposition of rust as shown in Fig 3.6.

The reactions may be represented as follows:



Other example: Ships which remain partially immersed in sea water for a long time undergo water line corrosion.

2. b. **Pitting Corrosion**

When a small dust particle gets deposited on a steel surface, the region below the dust particle is exposed to less oxygen compared to the remaining part. As a result, the region below the dust particle acts as anode and undergoes corrosion and forms a pit as shown in Fig: 3.7. The remaining region of the metal acts as cathode and is unaffected. Formation of a small anodic area and a large cathodic area results in intense corrosion below the dust particle.

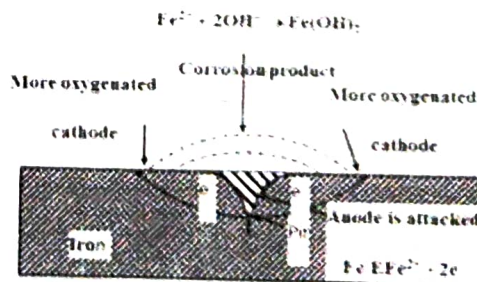
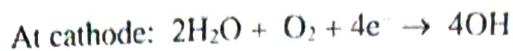
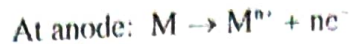


Fig: 3. 7 Pitting Corrosion

The reactions may be represented as follows:



Corrosion Control

Due to corrosion, there is a great loss of material and money. Therefore, it is essential to protect metals from corrosion. Since, there are two components involved in corrosion; the metal and environment both are considered in corrosion protection. Following methods have been adopted for the protection of metal from corrosion.

Galvanizing

Galvanizing is the process of coating a metal surface such as iron with zinc metal. Galvanizing of iron is an example of anodic metal coating on the surface of a cathodic metal.

Process: Galvanization is carried out by hot dipping method as shown in Fig.2.6. It involves the following steps

1. Solvent Cleaning: The metal surface is washed with organic solvents to remove organic impurities on the surface.
2. Alkali Cleaning: Residual organic impurities are removed by treating the object with alkali such as NaOH.
3. Pickling: Rust and Scale is removed by washing the object with dilute sulphuric acid H_2SO_4
4. Finally, the article is washed with water and air-dried.
5. The article is then dipped in a bath of molten zinc at $430^{\circ}C$ to $470^{\circ}C$ (Molten zinc is covered with a flux of ammonium chloride to prevent the oxidation of molten zinc.)
6. The excess zinc on the surface is removed by passing through a pair of hot rollers.

Application

Galvanization of iron is carried out to produce roofing sheets, fencing wire, buckets, bolts, nuts, pipes etc.

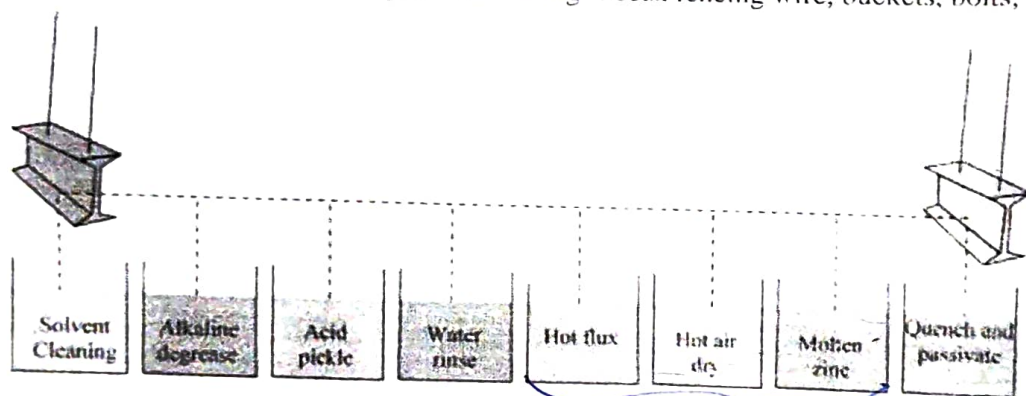


Fig: 3.8 Galvanizing

Galvanisation

Note: Even if the Zn coating falls off at some places, the base metal (Fe) does not get corroded at those places. (This is because the base metal acts as cathode. In corrosion process, the cathodic metal always remains unaffected.) And because of the formation of small cathodic area and larger anodic area corrosion process will not be there.

(**Note:** Galvanized articles are not used for preparing and storing food because zinc dissolves in dilute acids producing toxic zinc compounds)

M1
6.0 Anodizing

M2
6.6

(Anodizing of aluminium): It is a type of inorganic coating. A process in which top layer of a metal is converted to its oxide, which acts as a protective barrier and reduces corrosion is called as anodizing.

Anode is decreasing with polishing
Process: The article is subjected to degreasing and polishing and then made as anode, refer fig 16. Chromic acid electrolyte is maintained at about 35°C and suitable voltage is applied to get desired thickness of the oxide film. Anodised article is sealed by treating with boiling water or steam containing salts of Nickel or Cobalt. (The $\text{Al}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ layer on the surface acts as a protective coating, hence corrosion is prevented.)

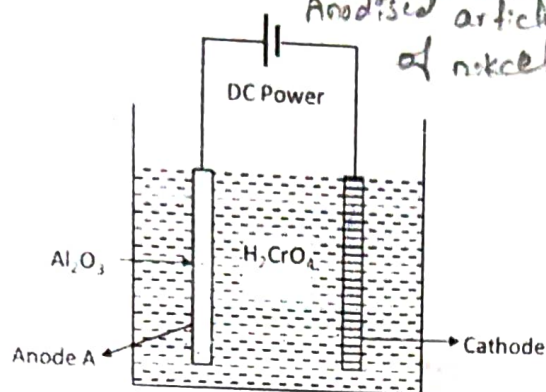
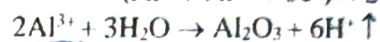
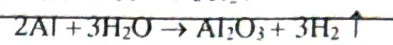


Fig: 3.9. Anodizing of aluminium

Anodised article is treating with boiling water & steam of nickel & cobalt
At Anode: $(\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-) \times 2$



At Cathode: $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2 \uparrow$



Other metals such as Mg, Ti etc. can also be anodized.

Applications: Metals such as Al, Mg, Ti etc. are anodized to control corrosion.

3.5.2 (b) Phosphating

It is generally obtained on steel surface by converting surface metal atoms into their phosphate by chemical or electrochemical process.

Phosphate bath contains three essential components 1. Free phosphoric acid 2. Primary phosphate like Fe, Mn, or Mg phosphate. 3. Accelerator such as nitrates, nitrites hydrogen peroxide and PH usually in the range of 1.8-3.2 if the process is immersion type. If the process is spraying type then PH is in an around 3.2 to 7.8 and temperature maintained at 35°C.

The mechanism of Phosphating involves following steps.

1. First dissolution of the metal as metal ions.
2. Metal ions reacting with phosphate ions to form a metal phosphate.
3. Deposition of the metal phosphate on the surface of the metal.

Phosphating is always given as underline (under coating) before the finishing, because Phosphating along with corrosion resistance it also imparts the surface a good paint adhesion quality.

Application: An undercoating before painting of automotive bodies, refrigerators, washing machines etc.

One of most important application of Phosphating is of galvanized iron which is otherwise difficult to paint.

3.6 Cathodic protection

A method of converting the metallic structure into cathode by supplying electrons from external source is called cathodic protection.

- i) A metallic structure is converted to cathode by connecting it to another highly reactive metal.

ii) A metallic structure is converted to cathode by supplying electrons DC source.

3.6.1 Sacrificial anode method:

In sacrificial anode method, the metal to be protected is electrically connected to a more active metal using insulated copper wire. For example, when steel is to be protected, it may be connected to a block of Mg or Zn. In such a situation, steel acts as **cathode** (high electrode potential) and is unaffected as shown Fig.3.10. Mg and Zn act as anode (low electrode potential) and undergo sacrificial corrosion. When the sacrificial anode gets exhausted, it is replaced with new ones.

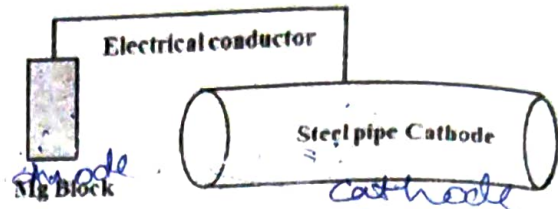


Fig: 3.10 Sacrificial anode technique

Other examples: Mg bars are fixed to the sides of ships to act as sacrificial anode.
Mg blocks are connected to underground pipe lines.
Mg blocks are connected to underground oil storage tank.)

3.6.2 Impressed current method:

In cathodic protection, the metal to be protected is completely converted into a cathode. Since cathodes do not undergo corrosion, the metal is protected against corrosion.

- In this method, an impressed current is applied in opposite direction to nullify the corrosion current, and convert the corroding metal from anode to cathode.
- The impressed current is slightly higher than the corrosion current. Thus the anodic corroding metal becomes cathodic and protected from corrosion.
- The impressed current is taken from a battery or rectified on A.C. line. The impressed current protection method is used for water tanks, water & oil pipe lines, transmission line towers etc.

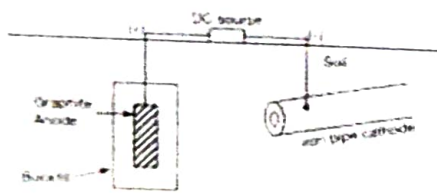


Fig: 3.11 Impressed Current Technique

In impressed current method, the metal to be protected is connected to the negative terminal of an external DC. Power supply. The positive terminal is connected to an inert electrode such as graphite. Under these conditions, the metal acts as cathode and hence does not undergo corrosion. The inert electrode acts as anode; but it does not undergo corrosion because it is inert refer Fig: 3.11.

3.7 Corrosion Penetration Rate (CPR)

Definition: The Corrosion penetration rate is the speed at which any metal or alloy deteriorates in a specific corrosive environment through chemical or electrochemical reactions.

It is also defined as the **amount of weight loss per year in the thickness of metal or alloy due to corrosion**. The Corrosion penetration rate also referred as corrosion rate.

Corrosion penetration rate depends on following factors;

1. Nature of metal.

2. Nature of corrosive environment/medium.
3. Nature of corrosion product.
4. Speed at which corrosion spreads in the inner portion of metal.

Imp (The Corrosion penetration rate is calculated using the following formula;)

$$CPR = \frac{k \times W}{D(\rho) \times A \times T}$$

Where,

W-is weight loss after exposure time

T-is exposure time in corrosive medium.

D- is the density of metal or alloy.

A-is surface area of exposed specimen.

K -is constant.

The CPR is conveniently expressed in terms of either mils (0.001 inch) per year (mpy) or milli meters per year (mmpy).

Handwritten conversions:

$$1 \text{ g} = 1000 \text{ mg}$$

$$1 \text{ m}^2 = 100 \times 100 \text{ cm}^2$$

$$1 \text{ m}^2 = 1550 \text{ inch}^2$$

$$1 \text{ inch}^2 = 6.45 \text{ cm}^2$$

$$1 \text{ cm}^2 = 0.155 \text{ inch}^2$$

	CPR in mpy	CPR mmpy
K	534	87.6
W	Mg	Mg
(D)ρ	g/cm ³	g/cm ³
A	inch ²	cm ²
Hrs	Hrs	Hrs

Numerical Problems on CPR

1. A sheet of carbon steel one meter wide by three meters long has lost 40g to corrosion over the past six months. Convert that mass loss to a penetration rate of the steel in mm units and mpy units. What would be the corrosion rate? (Carbon steel density=7.8g/ cm³) To calculate CPR in mmpy

Solution:

To calculate CPR in mmpy

	Given	CPR in mmpy
K		87.6
W	40 g	40 x 1000 mg
(D)ρ	7.8g/cm ³	7.8g/cm ³
A	1mx3m=3m ²	3x100x100cm ²
T	6 months	6 x30x24 hrs

$$CPR = \frac{k \times W}{D(\rho) \times A \times T} \quad \text{To calculate CPR in mpy}$$

	Given	CPR in mpy
K		534
W (wt loss)	40 g	40 x 1000 mg

Vapour Corrosion Inhibitors for Protecting Computer Circuit Boards:-

VCI - These are the chemicals that slowly release corrosion-inhibiting molecules.

* These vapours adsorb onto metal surface and form a protective molecular layer that prevents oxidation.

→ Circuit boards contain exposed copper, tin, Al & component leads that can benefit from VCI protection during storage.

Best VCI for electronics.

1) Nitrite-free / amine based VCI's.

Safer for Cu & Ag.

2) Electronics-Safe or Contact-Safe.

3) Polyfilms.

4) VCI foam inserts.

5) VCI emitter capsules.

used inside equipment during storage or shipment.

6) VCI paper.

Used for metal parts or electro-mechanical assemblies.

→ The compounds that VCI release into airspace actively prevent corrosion by forming strong bonds with a metal surface, maintaining the natural and healthy passive oxide layer on the metal.

→ The VCI layer that forms is invisible, dry and does not affect the physical properties or functionality of the metal in any way.

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10/11/17
12/11/17

* VCI can be added in small concentrations to prevent corrosion during testing, during wet storage and after testing. These inhibitors are to prevent corrosion of metallic components in 3 phases, within the soil at the waterline and above the waterline.

→ These inhibitors include aliphatic, aromatic, cyclo-hexylamines, aminonitrobenzoates, heteroalkylated lower amines, etc.

Anodic Inhibitors

- * Stifle the Corrosion reaction by forming a sparingly soluble compound with newly produced metal ion.
- * Forms thin passivating oxide film on the metallic surface.
- * Chromates, phosphates, tungstates or other transition element ion with high oxygen content.

Cathodic Inhibitors

- i) In an acidic solⁿ, the cathodic reaction is: $2H^+(aq) + 2e^- \rightarrow H_2(g)$
 - slowing down the diffusion of H^+ ion (amines, mercaptans, ureas, thiourea, etc).
 - Increasing the over voltage of H_2 evolution.
- ii) In neutral solⁿ, the cathodic reaction is: $H_2O(l) + \frac{1}{2}O_2(g) + 2e^- \rightarrow 2OH^-(aq)$
 - Eliminating Oxygen from corroding medium.
 - Retarding the diffusion to cathodic areas (Mg, Zn, Ni salts which react with OH^- ions).